

2026 AIO Training Exam I

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5 (and 2 halves) self-authored problems to practice for AIO 2026, all about Bob.

Sorry, too much Binary Search but maybe thats good so I'll add a training exam focused on a specific topic.

Bob is at Math Camp but is having lots of trouble because of the **first** practice problem, which asks him to find $(-1)^{2026}$. He is astonished to find that the answer to this problem is simply 1, and wants to investigate powers of -1.

You will be given one integer, n , the power of -1 you are to calculate.

Your output should consist of a single integer, $(-1)^n$.

5674

1

10000000000000000000000000000000

For all subtasks, $n \geq 0$.

2

Not Yummy Enough

Time Limit: 1.0 second **Input File:** Standard Input
Memory Limit: 32 MB **Output File:** Standard Output

After being at camp for so long, Bob has developed an intolerance to *non-yummy* food. However, he is only good at maths and needs your help to tell him how many meals he can still eat, given the foods in his house.

Input Format

The first line of input will contain two space-separated integer, n , and k , the number of meals available, and his food *intolerance level* (Bob will only eat food that has a strictly greater yumminess than his intolerance level), respectively.

The next line of input will contain n space separated integers, $a_1, a_2, \dots, a_n$, the available foods.

Output Format

Your output should consist of a single integer, the number of meals/foods Bob can eat.

Subtasks & Constraints

For all subtasks, $n \geq 0$, $0 \leq k \leq 10^6$, and for all i , $a_i \leq 10^6$.

Subtask	Points	Additional Constraints	Description
0	0	Bob is able to eat all of the food available	Trivial
1	1	$N \leq 1$	There is barely any food at home :(
2	99	$N \leq 10^6$	Bob has a lot of food in his house!

Nice Numbers

Time Limit: 1 seconds **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Bob thinks some numbers are more nice than others, but he can't exactly figure out why he looks at numbers and some just look better. After a lot of research, he finally thinks it has something to do with the frequency of certain digits in a constant size range after another specific digit.

To streamline his process of investigation, he has asked you to help him.

Input Format

The first line of input will contain 3 space-separated integers n , k , and m , the length of the number Bob is testing, the size range Bob suspects affects his perception of numbers, and the number of ranges he wants to test, respectively.

The second line of input contains an integer of length n , the number Bob is testing.

The next m lines of input contain 2 space-separated integers a_i and b_i , the start of each range of size k that Bob wants to investigate, and the frequency of digit b_i in that range.

Output Format

Your output should consist of m space-separated integers, the number of digit b_i in the interval from digit a_i to $a_i + k - 1$.

Subtasks & Constraints

For all subtasks and i , $n, k, m, a_i \geq 0$ and $b_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Subtask	Points	Additional Constraints	Description
1	5	$k = 1$	Bob only tests individual numbers
2	10	$m = 1$	Bob only tests 1 range
3	10	$a_i = a_j$ for all i and j	The start of each range is the same
4	15	$b_i = b_j$ for all i and j	You are looking for the same number
5	20	$n, m \leq 1000$	Bob is confirming his hypotheses
6	40	$n, m \leq 10^5$	Bob is dealing with big numbers

Numberline

Time Limit: 1.0 second **Input File:** Standard Input
Memory Limit: 1 GB **Output File:** Standard Output

Alice and Bob are playing a game of Numberline, where in each round, Alice chooses a number from the first a numbers of the number line, and Bob chooses a number from the first b numbers of the number line.

The winner is determined by who chooses a number closest to a number given to both players before they chooses.

Input Format

The first line of input contains two space-separated integers n and k , the number of numbers on the number-line, and the number of rounds both Alice and Bob will play, respectively. The next line of input contains n space-separated integers $a_1, a_2, \dots, a_n$, the numbers on the number line (a_1 is the first number, a_n is the last number).

The next k lines include an integer m_i , the number that players must be close to.

After this, the next k lines will include 2 space-separated integers x_i and y_i , where Alice chooses from the first x_i numbers and Bob chooses from the first y_i numbers.

Output Format

You should output k lines, one for each round. In each of these lines, you should print 'Alice wins' if Alice would win if both players played optimally, 'Bob wins' if Bob would win if both players played optimally, and 'Draw' if neither Alice or Bob would win if they both played optimally.

Subtasks & Constraints

For all subtasks, and for all i , $n, k, x_i, y_i, m_i \geq 0$, $k, a_i \leq 10^6$, and $x_i, y_i \leq n$

Subtask	Points	Additional Constraints	Description
1	10	$k = 1$ and $N \leq 10^3$	They play 1 round with a lot of numbers
2	15	$k = 1$ and $N \leq 10^6$	They play 1 round with loads of numbers
3	15	$N \leq 10^3$	They play with a lot of numbers
4	60	$N \leq 10^5$	They play loads of rounds with loads of numbers

AIC Disruption

Time Limit: 1.0 seconds **Input File:** Query Function
Memory Limit: 64 MB **Output File:** Reporting Function

Bob is passionately watching the AIC (Australian Icecream Cup) in 2027, but is suddenly captured by aliens and can't watch the World Cup. Since Bob is now a icecream superfan after failing miserably in the 2026 AIO because he didn't participate in the 2026 AIO Training Exam I by aperson31415, he really wants to know the standings of each country in the Icecream Cup.

However, the aliens are not very nice and have only given him Q_1 pieces of information, if team A survived more, less, or the same number of rounds as team B , and Q_2 pieces of information, if team A and B ever played each other. You can assume the Aliens are super smart and can answer your queries each in fast constant time.

Given that the number of teams is exactly a power of 2, and each team gets eliminated when they lose (there are no draws), Bob is desperate and wants you to help him find the team that beat team i for all teams i .

Input Format

You should call the function `teams()` $\rightarrow$ `int` to get the number of teams n . It is guaranteed that n is a power of 2.

Then, you should call the function `round_survived(teama: int, teamb: int)  $\rightarrow$  ['>', '<', '=']`, at most Q_1 times as described above.

You should call the function `played_eachother(teama: int, teamb: int)  $\rightarrow$  bool`, at most Q_2 times as described above.

Output Format

Once you have found (or you think you have found) the team that beat each team, you should call the function `report(beat: list[int])  $\rightarrow$  None` to the judge, where the i th element of `beat` is who beat the i th team. Note that you should output -1 for the winning team.

Subtasks & Constraints

For all subtasks, $n, Q_1, Q_2 \geq 0$.

Subtask	Points	Additional Constraints	Description
1	5	$N = 2, Q_1 \leq 2, Q_2 \leq 2$	There are only 2 teams
2	15	$N \leq 16, Q_1 \leq 256, Q_2 \leq 256$	The aliens are generous
3	20	$N \leq 1024, Q_1 \leq N^2, Q_2 \leq N \log_2 N$	Time hint, lots of teams
4	30	$N \leq 262144, Q_1 \leq N - 1, Q_2 \leq N \log_2 N$	Time hint, teams from every city
5	30	Functional Grading (See below)	Pretty much playing a game with them

For Subtask 5, $N \leq 262144$. Your score out of 30 will be calculated as $f(\sum Q)$ based on the number of queries used:

- If your program uses $Q_1 > N$ or $Q_2 > N \log_2 N$ at any point, you will receive 0 points for this subtask.
- If your program solves the testcase within the limits, your score will be determined by the formula:

$$f(\sum Q) = 30 \times \max\left(0, 1 - \frac{(Q_1 + Q_2) - 2N}{N \log_2 N}\right)$$

Speedrunning

Time Limit: 1.0 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Bob is super happy that the new AIO version 2026 (Asteroid Interception Operation) is being released soon, and as a result, has once again become addicted to speedrunning AIO (his best was an any 100% in 2 hours 10 minutes in version 2025). However, as the game is very new, there are many bugs in the game that Bob wants to exploit as soon as he can before they get patched.

The only way Bob can do a speedrun is to do the events in order, which each take 1 second, from start to finish, while exploiting bugs to skip over some non-essential events, which can take less or more than 1 second to exploit.

Input Format

The first line of input will contain three integers, n , m , and k , the number of 'events' that Bob has to complete, the number of bugs/skips that have been discovered, and the number of required 'event's that Bob needs to go through to still get a valid speedrun.

The next m lines of input contain three space-separated integers $e_{i,1}$, $e_{i,2}$, and m_i showing that a skip from event $e_{i,1}$ to $e_{i,2}$ is possible, which takes m_i seconds.

The next and last line of input contains k space-separated integers c_i , the events that Bob needs to complete to verify (essential, but not necessarily the only events that are completed).

Output Format

You should output one integer, the minimum number of seconds for Bob to finish the speedrun (get from first event to last event).

Subtasks & Constraints

For all subtasks and all i , $n, m, k, e_{i,1}, e_{i,2}, m_i, c_i \geq 0$, $e_{i,1} \leq e_{i,2}$, $e_{i,1}, e_{i,2} \leq n$, $c_i \leq n$

Additionally, no exploit is mentioned more than once, and no exploit skips the same events as another exploit.

Subtask	Points	Additional Constraints	Description
1	5	$m = 0$	There are no exploits
2	10	$k = 0, n \leq 10^3$	There are no essential events
3	10	$k = 0, n \leq 10^6$	There are no essential events, but a lot of events
4	20	$n, m, k \leq 10^3$	A normal sized game
5	55	$n, m, k \leq 10^5$	The game has too many exploits

Favourite Number

Time Limit: 1.0 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Bob has an array of integers. He also has a favourite number x . He can change a range/subarray of elements in the array by a certain number k times. He wants to find the most number of x 's in a row that he can make after applying his operations.

Input Format

The first line of input will contain 3 space-separated integers n , k , and x , the number of elements in the array, the number of operations he can do, and his favourite number, respectively. The next line of input will contain n integers, $a_1, a_2, \dots, a_n$, the numbers in the array.

Output Format

You should output one integer, the longest range of x 's that Bob can achieve.

Subtasks & Constraints

For all subtasks and all i , $n, k, x \geq 0$, $x, a_i \leq 10^6$.

Subtask	Points	Additional Constraints	Description
1	5	$k = 0$	You cannot do any operations
2	10	$k = 1, n \leq 10^6$	You can do 1 operation
3	15	$n, k \leq 100$	A normal sized array
4	20	$n, k \leq 2000$	A big array
5	50	$n, k \leq 10^5$	A huge array

Brief Editorials

Not meant to be comprehensive or understandable, did very quickly.

1 Powers of -1

Obviously *Powers of 2* is from selection exam, but I've heard of *Powers of 1* asw...

1. Just do by hand, if $N = 1$, output -1, if $N = 0$, output 1
2. Also do by hand, try to see some observations
3. Well you can multiply -1 by itself N times
4. Hopefully you've figured out N odd means -1, and N even means 1, just check the last digit

There is also a binary exponentiation but oh well.

2 Not Yummy Enough

Kinda me after every camp. Should be p1 instead I think.

0. Just yes because not enough subtasks
1. Just check if this number is $\geq k$
2. Just go through array and do it, or binary search and sort but that's slower

3 Nice Numbers

Could have used strings or int so that u cant just use a array or matrix. If u did maybe try using a frequency list.

1. Just check for each query if $n[a_i] = b_i$
2. Just go through the interval and count the occurrences
3. Build a frequency array or something like that for the interval
4. Do a sliding window and store values of how many occurrences of b_i
5. Loop through the intervals and complete search through them
6. Use a sliding window + frequency array, after sorting start of ranges (remember to output in correct order though)

Should have made less points for subtasks 2-5 I think.

4 Numberline

Originally more complex, but simplified. Also, I think the idea of being 'optimal' is very vague here.

1. For every pair of numbers in Alice and Bob's range's, check how close they are (for optimality), and who wins
2. You can first sort the numbers in both Alice and Bob's range's, then binary search for the closest above and below, and see which is closer and who wins
3. Do the strategy in previous subtask for all queries
4. To sort them, make it using something like a set so its amortised to $N \log N$ and not $N^2 \log N$, after sorting the range that Alice and Bob can access, and then same strategy as subtask 2

5 AIC Disruption

Why did I do this for AIO but its a nice introduction. Also, find out urself because I think its good to research on stuff not encountered yet. Then, pretty trivial with standard techniques. Not completely sure if 100 points is correct tho (70 definitely)?

6 Speedrunning

Kind of ambiguous terminology but wont affect I'm pretty sure.

1. Just sum up all the seconds, or just get n in $O(N)$, $O(1)$ respectively.
2. Do an easy dp
3. I don't really know how to do this without explaining most of 5, but make a dag pretty much and then dp
4. Also a simple dp but any skip that jumps over/skips a essential event u can just discard
5. Yeah, just a dag asw but use the same strategy in sub 4 to get rid of condition of essential events

7 Favourite Number

Seems a lot like a ds excercise so probably will be bashed. Maybe reducing mem lim will help.

1. Just loop through array and find biggest block
2. Go through array and find blocks of the same number. Then for each block, set all to k and see if there are any blocks of k touching
3. I don't know how you do this but maybe use strategy in sub2
4. Also just use strategy in sub3?
5. Use a 2 pointers or smth like that

Remarks

General

- Make a bunch of test cases and explanations
- Fix difficulty curve a bunch
- Check clarity and on yuantiji
- Too much binary search I think, but fine i can improve on later training exams

8 Powers of -1

- Kind of not AIO style more like atcoder: a bit too mathy
- Implementation would be kinda annoying like more than AIO 2020
- Maybe biased to people who know about this but pretty trivial even though that?

9 Not Yummy Enough

- Seems a bit too easy for p2 even though p2 is usually easy, maybe make harder?
- Quite trivial not much *problem solving*

10 Nice Numbers

- Kinda looks nice ig

11 Numberline

- $t_i = 3$ is much harder than other cases, maybe remove other cases
- Having too many choices/criteria makes implementation kinda annoying
- I feel like its a bit ambiguous

12 AIC Disruption

- Who ts is putting an interactive in an AIO training
- Functions are also ambiguous
- Subtasks kinda strange ig, improve

13 Speedrunning

- I feel like its a pretty nice problem
- General subtask information sucks a bunch
- Variables are named quite wierdly tho which contributes to subtask general info looking super wierd

14 Favourite Number

- Seems more like a p6 :(or data structure excercise
- Maybe make a better story to it
- Can u bash with some sort of segtree?